

ORION

PART 04

Satellite & Communication Layer

Satellite/NTN, terrestrial, mesh, edge and fallback connectivity

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Purpose

Abstract multiple communication technologies behind a common connectivity model so applications do not depend on one network path.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Connectivity model

Application → connectivity manager → fiber/cellular/NTN/edge adapters → routing and policy → destination.

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NTN fundamentals

Non-terrestrial networks use satellites or other aerial infrastructure. They introduce different latency, coverage, bandwidth, and availability characteristics.

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Path selection

Score paths using availability, latency, loss, bandwidth, cost, and policy. Make path recommendations explainable and auditable.

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Mesh and edge

Local nodes can keep limited services operating during upstream isolation. Store-and-forward supports intermittent links.

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Priority traffic

Emergency traffic can receive policy-defined priority, with bounded rules and auditability to reduce abuse.

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Security

Mutual authentication, encrypted transport, key rotation, device identity, replay protection, signed configuration, and field-device isolation.

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Testing

Simulate link loss, high latency, packet loss, gateway failure, route changes, and regional outages. Measure recovery and delivery success.

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Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

Satellite & Communication Layer should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.